

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua quaerat. Given a pair of two strings, . This problem can be restated as: what is the *longest unchanged part*, or *LUP*¹, between them?

Based on Myer's diff algorithm [Mye].

The most common use case of diffing is to compare source code, and to apply patches. For these use case, diffs are split per line². However, since our use case of diffing is markdown documents, and as confirmed through UX testing, we find it most useful to diff per word.

“Tung, Tuong, Bao.”

“Tung, Tuong, Khang.”

“Tung, Tuong, Bao.
Tung, Tuong, Khang.”

“Tung, Tuong, BaoKhang.”

And sometimes per sentence. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua quaerat.

“TTB-HCMUT is an indie developer group. We make experimental softwares.”

“TTB-HCMUT is an indie developer group. We research embedded softwares and AI applications.”

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The choice of diffing specificity is given to the user.

text
↓ string disassembling
tokens
↓ algorithm
changes

Fused diffing

text
↓
...
↓
changes

contextual disassembling
+ algorithm

Specialization

dedicated implementation as React component.

Parallelization

Utilizing fiber [Abs], we managed to offload the parallelize the running

Deep parallelization

can our Myer-based algorithm be optimized or changed to have run for branching paths?

[Mye] E. W. Myers, “An O(ND) Difference Algorithm and Its Variations.”

[Abs] in *Abstracting in-browser parallelism and more with fibers*, in bkhack developer handbook., TTB-HCMUT. [Online]. Available: <https://github.com/ttb-hcmut/bkhack/blob/main/src/Devbook/output/page-fiber.pdf>

¹also known in literature as *longest common substring* (LCS)

²speculation: a possible reason is that, historically, most (UNIX) tools work with textual data per line, for example `grep`.